



DUBAI MUNICIPALITY · WORLD'S FIRST AI-POWERED
PARK DESIGN CHALLENGE

ظلال الصفا

ZILAL AL SAFA

An AI-designed, self-cooling neighbourhood oasis for Al Safa
Park 2 — the first Dubai park *designed, simulated and operated*
by an AI comfort brain.

ZILAL AL SAFA

An AI-designed, self-cooling neighbourhood oasis for Al Safa Park 2

Dubai Municipality

AI Park Design Challenge

Plot 3570616 · 15,000 m²

Concept + Preliminary Design

ZILAL AL SAFA (Arabic *zīlāl* — “shades / canopies”) reimagines Al Safa Park 2 as a park that works **all day, all year**. Today the park is well-loved but the desert heat effectively closes it for roughly six hours a day in summer: usage collapses at midday and peaks only around 6 pm. Our single design goal is to **reclaim those lost hours** — using artificial intelligence, woven through the entire design process, to place every path, canopy and tree where the data says it does the most good.

18 → 72%

Reliable shade over
paths & activity zones

+5.5 h

Extra comfortable
hours/day, summer

−41%

Irrigation water demand

Net-positive

On-site solar energy

The idea — five moves

The park is organised around one native tree — the **Ghaf**, the UAE's national tree — and Emirati **majlis** culture, expressed through five connected moves:

1. **The Cool Loop** — a continuous, fully-shaded ~1 km walking/running/cycling circuit under solar-pergola canopies, its geometry AI-optimised to stay in shade at 3 pm in July.
2. **The Majlis Grove** — the social heart: clustered shaded majlis rooms under a grove of Ghaf & Sidr trees, placed in the coolest core and tuned to the 6 pm peak.
3. **Adaptive Play & Sport** — the existing football pitch and basketball court retained and shaded, plus an all-abilities, sensory playscape designed with the neighbouring health centre and school in mind.
4. **The Sponge Garden** — xeriscape planting, bioswales and AC-condensate + greywater reuse that cut irrigation demand while retaining and augmenting the park's mature trees.
5. **The Solar Canopy Spine** — photovoltaic pergolas that shade by day and power lighting, misting and the park app by night, making the park energy net-positive.

Supporting these: four distinctive **identity gateways** (one addressing each neighbour), a food & craft **market and café**, prayer space, restrooms, a dog run, and a year-round **events calendar**. On top sits a stack of **four benchmarked Dubai-first innovations** — the AI Comfort Brain, a sun-tracking kinetic mashrabiya canopy, a net-positive solar canopy and an interactive “Every Step Counts” kinetic floor — making the park genuinely interactive and **unique on every side** (see over).

A complete, compliant programme

The design delivers every element of the required brief: a ~1 km jogging track, inclusive play, retained sport, community & picnic space, native landscape, shaded seating, restrooms, drinking fountains, café/kiosks, prayer space, waste & recycling, bicycle parking, accessible drop-off, four entrances, and a Commercial & Service Facilities plan sized to ~15% leasable area – placed to activate the park at the evening peak and to fund its operation.

Artificial intelligence — the differentiator

The brief does not ask for AI-made pictures; it asks for AI integrated across analysis, ideation, development and decision-making, with human creativity and judgement retained. ZILAL is built on a **documented, reproducible six-stage pipeline** — **Ingest** → **Generate** → **Simulate** → **Forecast** → **Visualise** → **Operate** — in which AI expands options and quantifies trade-offs at every step, and the design team makes every final spatial decision. Uniquely, the pipeline does not stop at the drawing: the sixth stage is a live **Comfort Brain** digital twin that **operates the built park** — routing visitors to real-time comfort and auto-tuning misting, irrigation and lighting. No Dubai park does this. The full method is set out in the accompanying *AI Methodology Report*.

Feasibility — buildable within AED 35 million

At roughly **AED 2,300 / m²** across the 15,000 m² site, ZILAL is scoped to the stated budget by reusing what works (the existing pitch, court and mature trees), building in three phases, and letting commerce fund operations. Kiosks, café and events generate revenue that offsets O&M, while xeriscape planting and on-site solar cut running costs.

PHASE	FOCUS	DELIVERS
1 — Comfort first	Cool Loop, Majlis Grove core, entrances & universal access	Shade and daily usability, fast
2 — Play, sport & water	Adaptive play, shaded courts & fitness, Sponge Garden, restrooms & kiosks	Inclusive activity & water resilience
3 — Energy, AI & interactive	Solar Canopy Spine, the Comfort Brain & sensors, kinetic mashrabiya, footstep-energy hero zones, AR grove	Net-positive energy & a genuinely interactive smart park

How the proposal scores

ZILAL AL SAFA

Built to score on the official jury matrix

ZILAL is engineered to be strongest exactly where the jury weight sits – around **80% of the score** is Innovation, Human-centred sustainability, AI integration and Presentation.

JURY CRITERION	WEIGHT	ZILAL'S STRENGTH
Innovation & Creativity	20%	Four benchmarked Dubai-firsts – the AI Comfort Brain, kinetic mashrabiya, net-positive solar canopy & “every step counts” kinetic floor – around a Ghaf identity
Human-Centred Design, Sustainability & Quality of Life	20%	Five audiences + People of Determination; -41% water; net-positive energy; +5.5 comfortable hrs/day
Integration & Effective Use of AI	20%	Documented six-stage pipeline that also operates the park (the Comfort Brain); named tools, alternatives tested, human-led
Quality of Presentation & Communication	20%	Interactive microsite + boards, data visualisations, day & night renders, 1-minute animation
Feasibility & Implementation	15%	Scoped to AED 35M; phased; reuses courts & trees; commerce funds operations
Quality of Design & User Experience	5%	Legible active-edge / quiet-core zoning; seamless day-to-night activation

In one line: ZILAL AL SAFA turns a park the heat closes for half the day into a year-round, AI-designed, self-cooling oasis for the whole community – innovative, human-centred, sustainable and buildable within budget.

All performance figures are modelled/illustrative estimates for concept communication, grounded in Dubai Municipality Neighborhood Parks Manual targets, regional climate norms and public site data; to be firmed up with site survey and quantity-surveyor costing. This summary accompanies the full ZILAL AL SAFA submission: interactive microsite, drawing set, board set, renders, data charts, AI Methodology Report and one-minute concept animation.